# Michigan Tech

# Periodic Orbits and Winding Numbers

Allan Struthers
Math Sciences

MA4535 — Nonlinear Dynamics & Chaos | Topic notes | September 27, 2026

# Contents

Periodic Orbits, Newton Systems, and Stability

# Periodic Orbits and Winding Numbers

## Periodicity and first return

Let $\Sigma^+$ be a transverse section with one chosen crossing direction.

- The **First Return Time** $\tau(\boldsymbol{q})$ is the first positive time that the trajectory from $\boldsymbol{q} \in \Sigma^+$ returns to $\Sigma^+$.
- The **First Return Map** records that next crossing:

$$\boldsymbol{p}(\boldsymbol{q}) = \boldsymbol{y}\big(\tau(\boldsymbol{q}); \boldsymbol{q}\big), \qquad \boldsymbol{q}_{j+1} = \boldsymbol{p}(\boldsymbol{q}_j).$$

- A periodic orbit closes in the flow. Its crossings form a periodic orbit of the map $\boldsymbol{p}$.

## Once-around periodic orbit

Choose $\Sigma^+$ so the Rössler spiral crosses it once per turn.

- A once-around periodic orbit returns to the same section point after one turn:

$$\boldsymbol{p}(\boldsymbol{q}) = \boldsymbol{q}.$$

- Define the fixed-point residual

$$\boldsymbol{r}(\boldsymbol{q}) = \boldsymbol{p}(\boldsymbol{q}) - \boldsymbol{q}.$$

  The periodic orbit problem is $\boldsymbol{r}(\boldsymbol{q}) = \boldsymbol{0}$.

- This is a nonlinear equation on the section. We solve it with Newton's method, starting from a nearly returning trajectory.

## Newton step from the sensitivity equation

Integrate the trajectory and its sensitivity matrix together:

$$\boldsymbol{y}' = \boldsymbol{f}(\boldsymbol{y}), \qquad \boldsymbol{S}' = D\boldsymbol{f}(\boldsymbol{y})\boldsymbol{S}, \qquad \boldsymbol{S}(0) = I.$$

- At the first return, $\boldsymbol{S}(\tau)$ gives sensitivity to the starting point. Correct it for the changing return time to obtain $D\boldsymbol{p}(\boldsymbol{q})$ on $\Sigma^+$.
- Since $D\boldsymbol{r}(\boldsymbol{q}) = D\boldsymbol{p}(\boldsymbol{q}) - I$, one Newton step is

$$\big(D\boldsymbol{p}(\boldsymbol{q}^{(\ell)}) - I\big)\Delta\boldsymbol{q} = -\boldsymbol{r}(\boldsymbol{q}^{(\ell)}), \qquad \boldsymbol{q}^{(\ell+1)} = \boldsymbol{q}^{(\ell)} + \Delta\boldsymbol{q}.$$

- Thus the same sensitivity equation used earlier supplies the derivative needed by Newton's method.

## Twice around with an intermediate state

Search for both section crossings $\boldsymbol{q}_0$ and $\boldsymbol{q}_1$. Stack them into one unknown:

$$\boldsymbol{z} = \begin{pmatrix} \boldsymbol{q}_0 \\ \boldsymbol{q}_1 \end{pmatrix}$$

One return must carry each point to the other:

$$\boldsymbol{p}(\boldsymbol{q}_0) = \boldsymbol{q}_1, \qquad \boldsymbol{p}(\boldsymbol{q}_1) = \boldsymbol{q}_0$$

Define the coupled residual

$$\boldsymbol{r}_2(\boldsymbol{z}) = \begin{pmatrix} \boldsymbol{p}(\boldsymbol{q}_0) - \boldsymbol{q}_1 \\ \boldsymbol{p}(\boldsymbol{q}_1) - \boldsymbol{q}_0 \end{pmatrix} = \boldsymbol{0}$$

- This system searches for the intermediate state instead of hiding it inside $\boldsymbol{p}^2$.
- Require $\boldsymbol{q}_0 \neq \boldsymbol{q}_1$ to exclude a once-around fixed point.

## Block Newton system for two returns

At Newton iteration $\ell$, define the stacked correction

$$\Delta \boldsymbol{z}^{(\ell)} = \begin{pmatrix} \Delta \boldsymbol{q}_0^{(\ell)} \\ \Delta \boldsymbol{q}_1^{(\ell)} \end{pmatrix}$$

Compute $\boldsymbol{A}_j = D\boldsymbol{p}(\boldsymbol{q}_j^{(\ell)})$ from the sensitivity equation on each return. Then solve

$$\boldsymbol{J}_2 \Delta \boldsymbol{z}^{(\ell)} = -\boldsymbol{r}_2(\boldsymbol{z}^{(\ell)}), \qquad \boldsymbol{J}_2 = \begin{pmatrix} \boldsymbol{A}_0 & -I \\ -I & \boldsymbol{A}_1 \end{pmatrix}$$

- Update both crossings together: $\boldsymbol{z}^{(\ell+1)} = \boldsymbol{z}^{(\ell)} + \Delta \boldsymbol{z}^{(\ell)}$.
- Forming $\boldsymbol{A}_1 \boldsymbol{A}_0$ compounds expansion and contraction over both returns. Its condition number can be much larger than those of the one-return sensitivities.
- The block system keeps the two legs separate and is usually better conditioned.

# Three times around: target $r_3(z_3) = 0$

To exclude a once-around fixed point, require $q_0 \neq q_1$. Three is prime, so the remaining cycle has minimal map period 3.

Stacked state and residual

$$z_3 = \begin{pmatrix} q_0 \\ q_1 \\ q_2 \end{pmatrix}$$

$$r_3(z_3) = \begin{pmatrix} p(q_0) - q_1 \\ p(q_1) - q_2 \\ p(q_2) - q_0 \end{pmatrix}$$

Block Newton system

$$A_j = Dp(q_j)$$

$$J_3 = \begin{pmatrix} A_0 & -I & 0 \\ 0 & A_1 & -I \\ -I & 0 & A_2 \end{pmatrix}$$

$$J_3 \Delta z_3 = -r_3(z_3)$$

## Four times around: target $r_4(z_4) = 0$

To exclude once-around and twice-around cycles, require $\boldsymbol{q}_0 \neq \boldsymbol{q}_1$ and $\boldsymbol{q}_0 \neq \boldsymbol{q}_2$, respectively. The remaining cycle has minimal map period 4.

Stacked state and residual

$$z_4 = \begin{pmatrix} \boldsymbol{q}_0 \\ \boldsymbol{q}_1 \\ \boldsymbol{q}_2 \\ \boldsymbol{q}_3 \end{pmatrix}$$

$$r_4(z_4) = \begin{pmatrix} \boldsymbol{p}(\boldsymbol{q}_0) - \boldsymbol{q}_1 \\ \boldsymbol{p}(\boldsymbol{q}_1) - \boldsymbol{q}_2 \\ \boldsymbol{p}(\boldsymbol{q}_2) - \boldsymbol{q}_3 \\ \boldsymbol{p}(\boldsymbol{q}_3) - \boldsymbol{q}_0 \end{pmatrix}$$

Block Newton system

$$\boldsymbol{A}_j = D\boldsymbol{p}(\boldsymbol{q}_j)$$

$$J_4 = \begin{pmatrix} \boldsymbol{A}_0 & -I & 0 & 0 \\ 0 & \boldsymbol{A}_1 & -I & 0 \\ 0 & 0 & \boldsymbol{A}_2 & -I \\ -I & 0 & 0 & \boldsymbol{A}_3 \end{pmatrix}$$

$$J_4 \Delta z_4 = -r_4(z_4)$$

# $m$ times around: target $r_m(z_m) = 0$

To obtain minimal map period $m$, require $\boldsymbol{q}_d \neq \boldsymbol{q}_0$ for every proper divisor $d$ of $m$.

Stacked state and residual

$$\boldsymbol{z}_m = \begin{pmatrix} \boldsymbol{q}_0 \\ \boldsymbol{q}_1 \\ \vdots \\ \boldsymbol{q}_{m-1} \end{pmatrix}$$

$$\boldsymbol{r}_m(\boldsymbol{z}_m) = \begin{pmatrix} \boldsymbol{p}(\boldsymbol{q}_0) - \boldsymbol{q}_1 \\ \boldsymbol{p}(\boldsymbol{q}_1) - \boldsymbol{q}_2 \\ \vdots \\ \boldsymbol{p}(\boldsymbol{q}_{m-1}) - \boldsymbol{q}_0 \end{pmatrix}$$

Block Newton system

$$\boldsymbol{A}_j = D\boldsymbol{p}(\boldsymbol{q}_j)$$

$$\boldsymbol{J}_m = \begin{pmatrix} \boldsymbol{A}_0 & -I & 0 & \cdots & 0 \\ 0 & \boldsymbol{A}_1 & -I & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \boldsymbol{A}_{m-2} & -I \\ -I & 0 & \cdots & 0 & \boldsymbol{A}_{m-1} \end{pmatrix}$$

$$\boldsymbol{J}_m \Delta \boldsymbol{z}_m = -\boldsymbol{r}_m(\boldsymbol{z}_m)$$

## Multiple returns

The $m$th return is the composition

$$\boldsymbol{p}^m = \underbrace{\boldsymbol{p} \circ \boldsymbol{p} \circ \cdots \circ \boldsymbol{p}}_{m \text{ times}}.$$

- A point has **Minimal Map Period** $m$ when

$$\boldsymbol{p}^m(\boldsymbol{q}) = \boldsymbol{q}, \qquad \boldsymbol{p}^j(\boldsymbol{q}) \neq \boldsymbol{q} \quad (1 \leq j < m).$$

- The corresponding flow orbit closes after $m$ selected crossings. Its elapsed period is

$$T = \sum_{j=0}^{m-1} \tau\big(\boldsymbol{p}^j(\boldsymbol{q})\big).$$

- $m$ counts returns. $T$ measures time.

## Turns around the Rössler spiral

Choose a vertical half-plane $\Sigma^+$ through the origin and select one crossing direction.

- A trajectory near the familiar Rössler spiral crosses this half-plane once on each trip around the spiral.
- Here "winding number" means the number of these turns before the periodic orbit closes. We use the term only for this picture.
- A fixed point of $\boldsymbol{p}$ goes around once. A point of minimal map period $m$ goes around $m$ times.
- Minimal map periods $m = 2, 4, 8$ therefore give two-, four-, and eight-turn Rössler orbits.

## Periodic orbits as nonlinear equations

For a parameter $c$, define the periodic-point residual

$$\boldsymbol{r}_m(\boldsymbol{q}, c) = \boldsymbol{p}_c^m(\boldsymbol{q}) - \boldsymbol{q}.$$

- Solve $\boldsymbol{r}_m(\boldsymbol{q}, c) = \boldsymbol{0}$ on $\Sigma^+$.
- Reject roots of $\boldsymbol{r}_j = \boldsymbol{0}$ for divisors $j < m$. This enforces minimal map period $m$.
- Recover every crossing from $\boldsymbol{q}_j = \boldsymbol{p}_c^j(\boldsymbol{q})$, $j = 0, \ldots, m-1$.
- Integrate between consecutive crossings to reconstruct the periodic orbit in the full Rössler state space.

## Computing the 2-, 4-, and 8-turn orbits

1: Choose $m \in \{2, 4, 8\}$ and a parameter value $c$
2: Use section data to choose an initial guess $\boldsymbol{q}^{(0)}$
3: **for** $\ell = 0, 1, 2, \ldots$ **do**
4: Evaluate $\boldsymbol{p}_c^m(\boldsymbol{q}^{(\ell)})$ by event-driven integration
5: Form $\boldsymbol{r}_m = \boldsymbol{p}_c^m(\boldsymbol{q}^{(\ell)}) - \boldsymbol{q}^{(\ell)}$
6: Solve $D\boldsymbol{r}_m \, \Delta\boldsymbol{q} = -\boldsymbol{r}_m$
7: $\boldsymbol{q}^{(\ell+1)} \leftarrow \boldsymbol{q}^{(\ell)} + \Delta\boldsymbol{q}$
8: **end for**

Continue the solution with minimal map period 2 in $c$, then use each doubling to initialize the solves for minimal map periods 4 and 8.

## Stability on the section

At a periodic point $\boldsymbol{q}$ with minimal map period $m$, linearize the $m$th return:

$$D(\boldsymbol{p}_c^m)(\boldsymbol{q}) = D\boldsymbol{p}_c(\boldsymbol{q}_{m-1}) \cdots D\boldsymbol{p}_c(\boldsymbol{q}_1) D\boldsymbol{p}_c(\boldsymbol{q}_0).$$

- Its eigenvalues are the multipliers of the periodic point.
- The orbit is attracting on the section when every multiplier has magnitude less than one.
- The section removes the neutral flow direction. We test the dynamics transverse to the periodic orbit.

## Period doubling

Suppose an orbit of the return map with minimal map period $m$ varies with the parameter $c$.

- A **Period-Doubling Bifurcation** occurs when a multiplier of $D(\boldsymbol{p}_c^m)$ passes through $-1$.
- The original orbit with minimal map period $m$ changes stability. A nearby orbit of the return map with minimal map period $2m$ appears under the usual nondegeneracy conditions.
- Repeated doublings produce the sequence

$$m = 1 \longrightarrow 2 \longrightarrow 4 \longrightarrow 8 \longrightarrow \cdots .$$

- Numerically, continue each branch in $c$ and monitor its multipliers. Use the doubled crossing pattern as the next guess.

# References I

[1] Yanghong Huang.
Math 44041/64041 applied dynamical systems: Linearisation and nonlinear terms.
University of Manchester course notes, 2020.

[2] Yanghong Huang.
Math 44041/64041 applied dynamical systems: Stability and lyapunov functions.
University of Manchester course notes, 2020.

[3] Yuri A. Kuznetsov.
*Elements of Applied Bifurcation Theory*, volume 112 of *Applied Mathematical Sciences*.
Springer, 3 edition, 2004.

# References II

[4] Semyon Malykh, Yuliya Bakhanova, Alexey Kazakov, Krishna Pusuluri, and Andrey Shilnikov.
Homoclinic chaos in the rössler model.
*Chaos*, 30(11):113126, 2020.

[5] Otto E. Rössler.
An equation for continuous chaos.
*Physics Letters A*, 57(5):397–398, 1976.

[6] Leonid P. Shilnikov and Andrey Shilnikov.
Shilnikov bifurcation.
*Scholarpedia*, 2(8):1891, 2007.